Azure Data Lakehouse Implementation: Medallion Architecture Engineering

Executive Summary **By Marra Mohamed**

The transition to a Data Lakehouse architecture on Microsoft Azure represents a strategic effort to process massive data volumes with high velocity and reliability. This project implements a modern data stack designed to ingest heterogeneous data from relational databases (Azure SQL) and REST APIs, centralizing them in a secure storage environment.

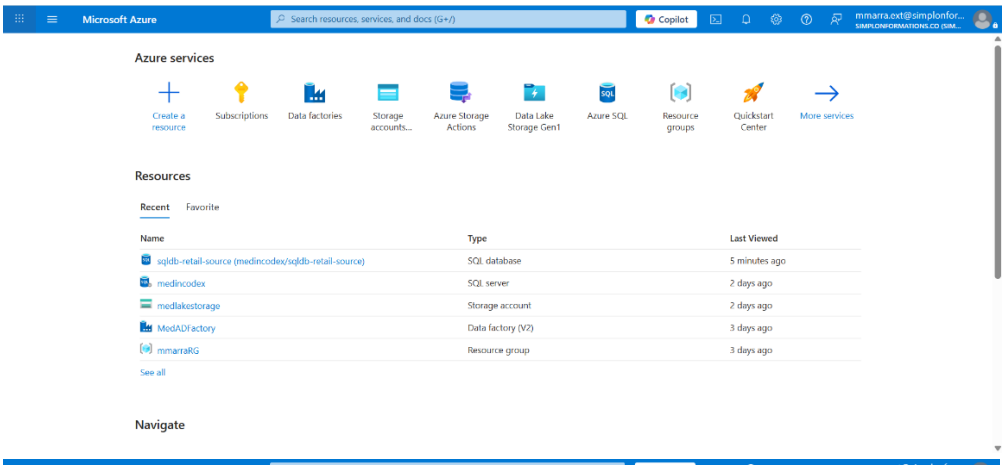
The core of the strategy is the **Medallion Architecture** (comprising Bronze, Silver, and Gold layers), which ensures total data traceability and high quality. Key outcomes include the establishment of a "Single Source of Truth" in the Silver layer and the creation of high-performance, aggregated "One Big Table" (OBT) models in the Gold layer for Business Intelligence (BI) consumption. The architecture achieves scalability by decoupling storage (Azure Data Lake Storage Gen2) from compute (Databricks), reducing "Time-to-Insight" through parallelized orchestration in Azure Data Factory.

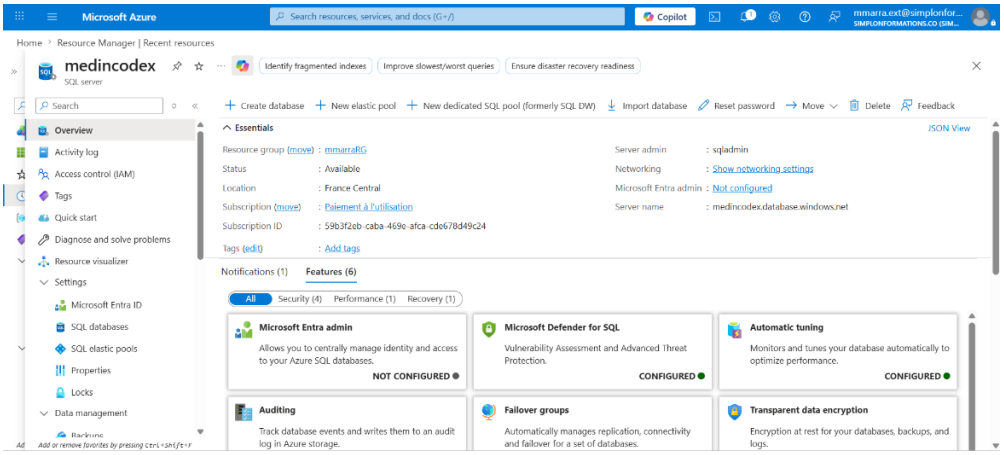
Technical Infrastructure and Tooling

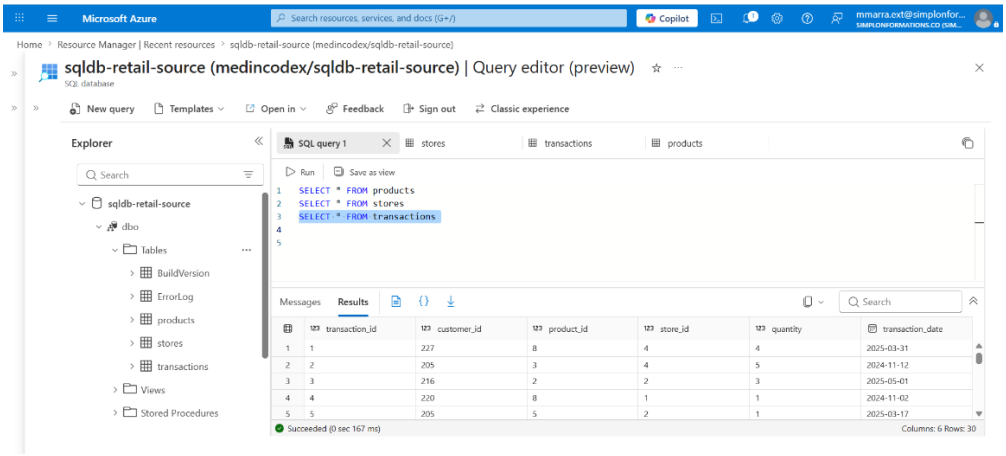
The architecture is built exclusively on Azure Cloud-native services, selected for their interoperability and security.

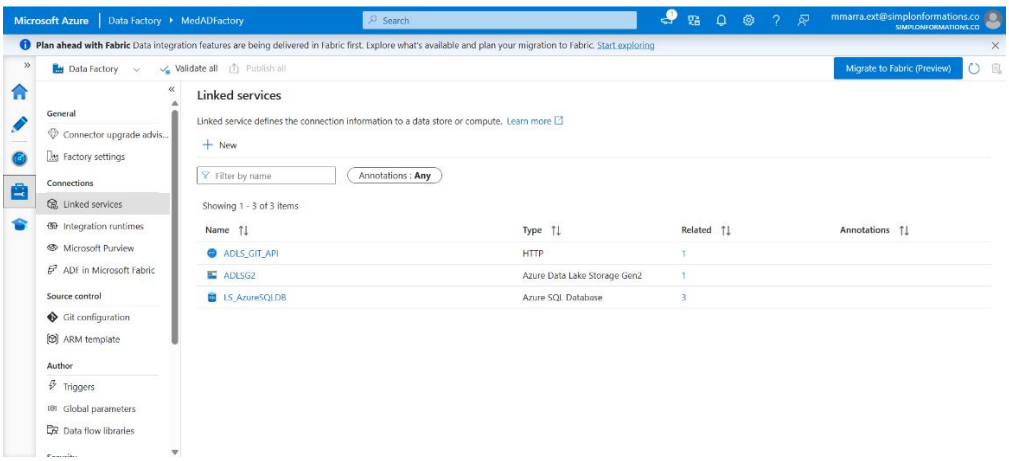
Core Components

Component	Role	Specific Usage
Azure SQL Database	Source (OLTP)	Simulates transactional systems (ERP/CRM) containing Products, Stores, and Transactions.
External REST API	Source (JSON)	Provides semi-structured Customer data for ingestion.
ADLS Gen2	Storage Foundation	Hierarchical storage optimized for Big Data analytics, secured via Access Keys.
Azure Data Factory (ADF)	Orchestrator	Manages ETL/ELT schedules, triggers, and pipeline monitoring.
Databricks (Apache Spark)	Compute Engine	Executes PySpark scripts for distributed data transformation and Delta Lake management.

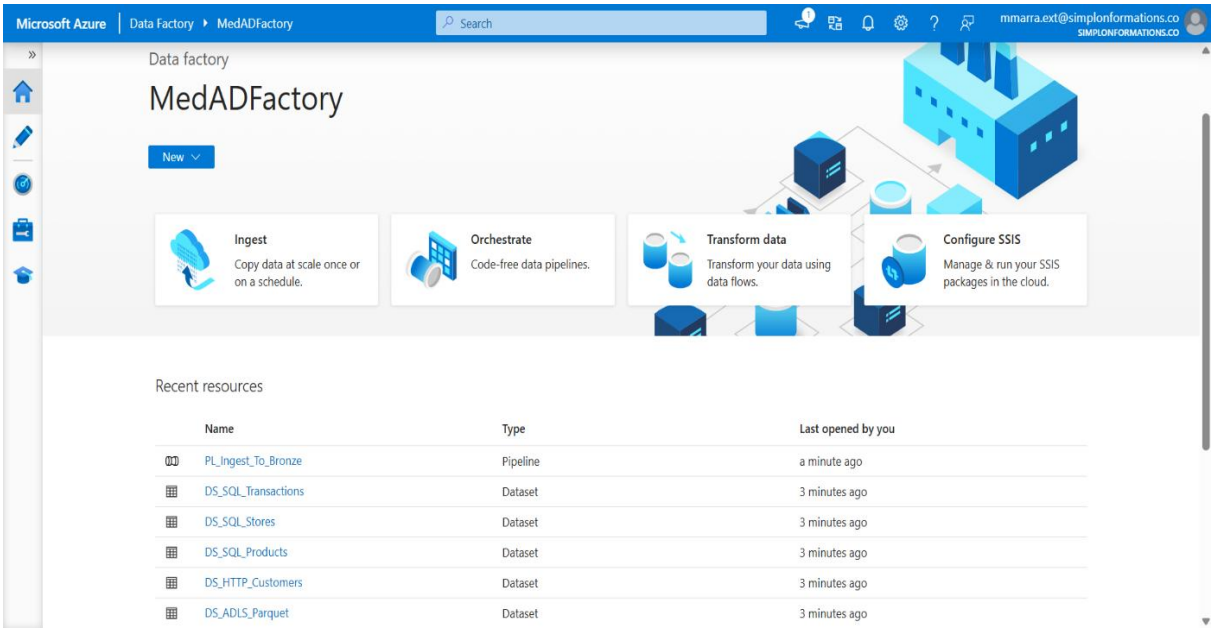

Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
mmarra.ext@simplonfor...
SIMPLONFORMATIONS.CO (SIM...
Azure services
Create a resource
Subscriptions
Data factories
Storage accounts...
Azure Storage Actions
Data Lake Storage Gen1
Azure SQL
Resource groups
Quickstart Center
More services
Resources
Recent
Favorite
Name
Type
Last Viewed
sqldb-retail-source (medincodex/sqldb-retail-source)
SQL database
5 minutes ago
medincodex
SQL server
2 days ago
medlakestorage
Storage account
2 days ago
MedADFactory
Data factory (V2)
3 days ago
mmarraRG
Resource group
3 days ago
See all
Navigate

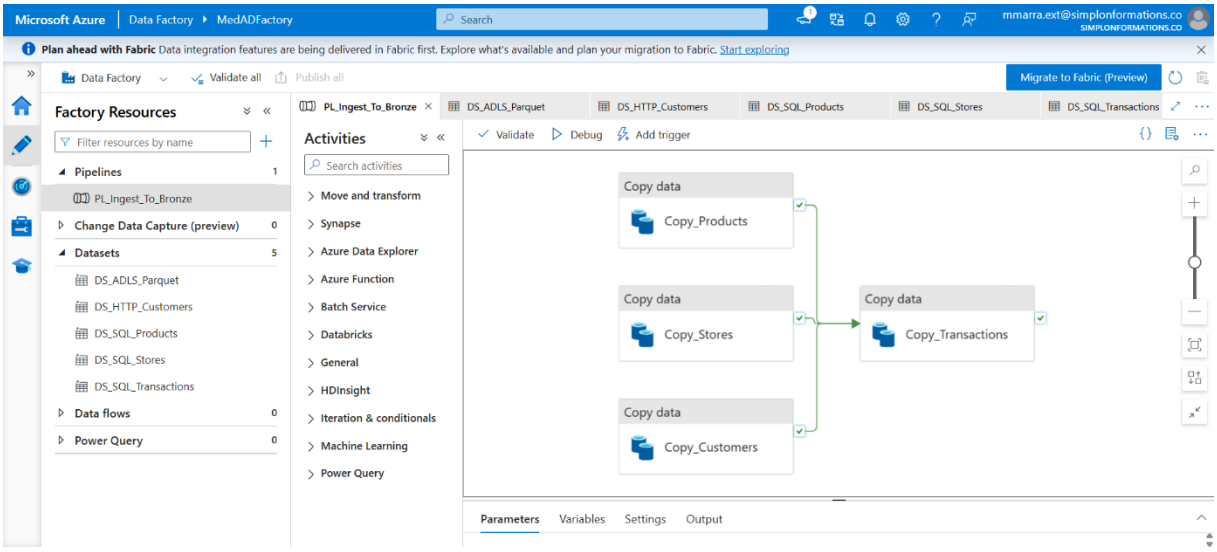

Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
mmarra.ext@simplonfor...
SIMPLONFORMATIONS.CO (SIM...
Home
Resource Manager | Recent resources
medincodex
SQL server
Identify fragmented indexes
Improve slowest/worst queries
Ensure disaster recovery readiness
Create database
New elastic pool
New dedicated SQL pool (formerly SQL DW)
Import database
Reset password
Move
Delete
Feedback
Overview
Activity log
Access control (IAM)
Tags
Quick start
Diagnose and solve problems
Resource visualizer
Settings
Microsoft Entra ID
SQL databases
SQL elastic pools
Properties
Locks
Data management
Essentials
JSON View
Resource group (move) : mmarraRG
Status : Available
Location : France Central
Subscription (move) : Paiement à l'utilisation
Subscription ID : 59b3f2eb-caba-469e-afca-cde678d49c24
Tags (edit) : Add tags
Server admin : sqladmin
Networking : Show networking settings
Microsoft Entra admin : Not configured
Server name : medincodex.database.windows.net
Notifications (1)
Features (6)
All
Security (4)
Performance (1)
Recovery (1)
Microsoft Entra admin
Allows you to centrally manage identity and access to your Azure SQL databases.
NOT CONFIGURED
Microsoft Defender for SQL
Vulnerability Assessment and Advanced Threat Protection.
CONFIGURED
Automatic tuning
Monitors and tunes your database automatically to optimize performance.
CONFIGURED
Auditing
Track database events and writes them to an audit log in Azure storage.
Failover groups
Automatically manages replication, connectivity and failover for a set of databases.
Transparent data encryption
Encryption at rest for your databases, backups, and logs.

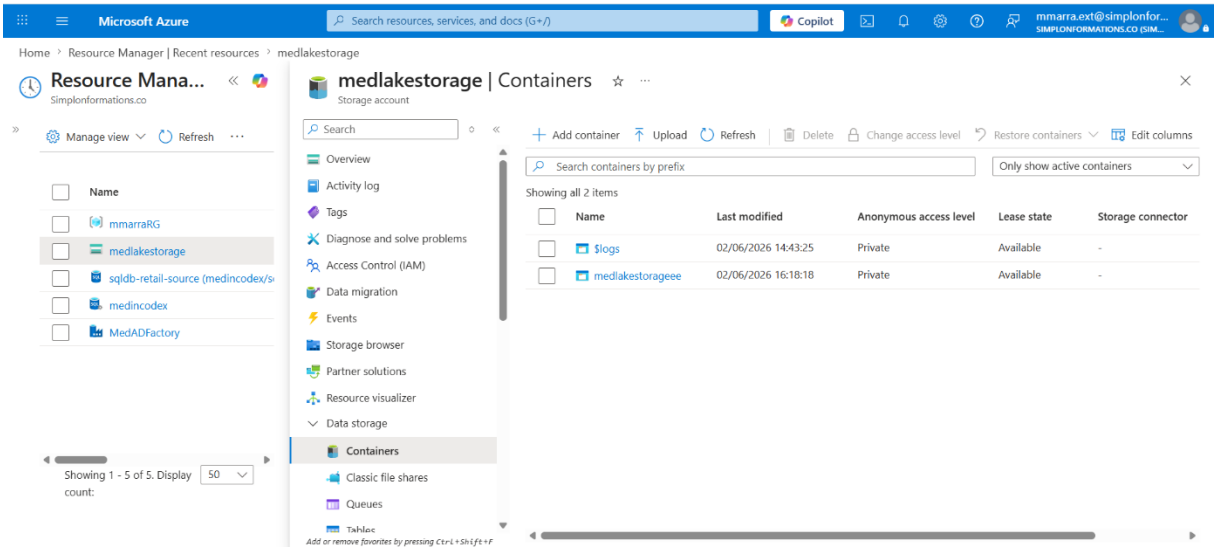

Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
mmarra.ext@simplonfor...
SIMPLONFORMATIONS.CO (SIM...
Home
Resource Manager | Recent resources
sqldb-retail-source (medincodex/sqldb-retail-source)
sqldb-retail-source (medincodex/sqldb-retail-source) | Query editor (preview)
SQL database
New query
Templates
Open in
Feedback
Sign out
Classic experience
Explorer
Search
sqldb-retail-source
dbo
Tables
BuildVersion
ErrorLog
products
stores
transactions
Views
Stored Procedures
SQL query 1
stores
transactions
products
Run
Save as view
SELECT * FROM products
SELECT * FROM stores
SELECT * FROM transactions
Messages
Results
Search
transaction_id
customer_id
product_id
store_id
quantity
transaction_date
1 1 227 8 4 4 2025-03-31
2 2 205 3 4 5 2024-11-12
3 3 216 2 2 3 2025-05-01
4 4 220 8 1 1 2024-11-02
5 5 205 5 2 1 2025-03-17
Succeeded (0 sec 167 ms)
Columns: 6 Rows: 30


Microsoft Azure
Data Factory
MedADFactory
Search
mmarra.ext@simplonformations.co
SIMPLONFORMATIONS.CO
Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. Start exploring
Data Factory
Validate all
Publish all
Migrate to Fabric (Preview)
General
Connector upgrade advis...
Factory settings
Connections
Linked services
Integration runtimes
Microsoft Purview
ADF in Microsoft Fabric
Source control
Git configuration
ARM template
Author
Triggers
Global parameters
Data flow libraries
Linked services
Linked service defines the connection information to a data store or compute. Learn more
New
Filter by name
Annotations : Any
Showing 1 - 3 of 3 items
Name
Type
Related
Annotations
ADLS_GIT_API
HTTP
1
ADLSG2
Azure Data Lake Storage Gen2
1
LS_AzureSQLDB
Azure SQL Database
3

Phase I: Raw Data Ingestion (Bronze Layer)







The Bronze layer's primary objective is to capture the complete history of source data in its raw state without any business-logic alterations. This allows for data reprocessing or "replaying" transformations if required.

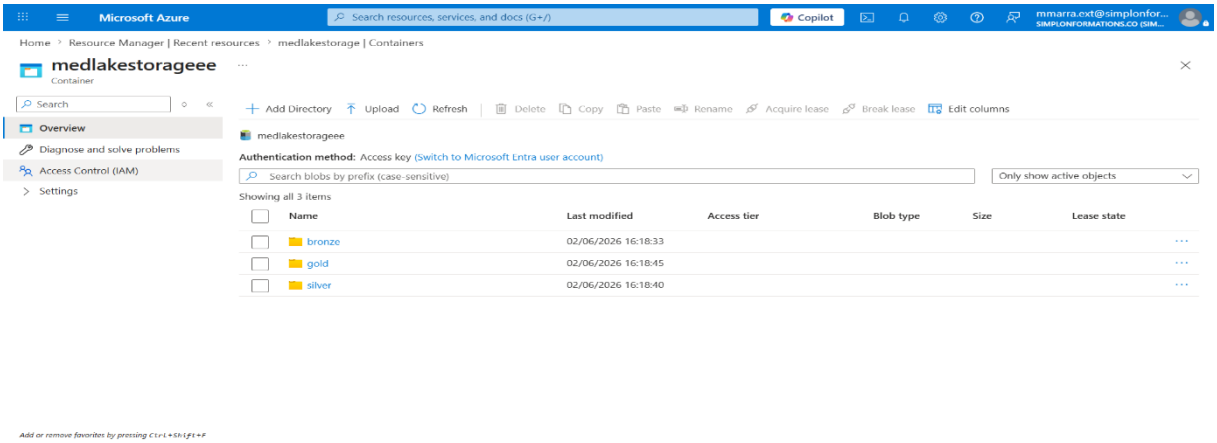
Connection Security

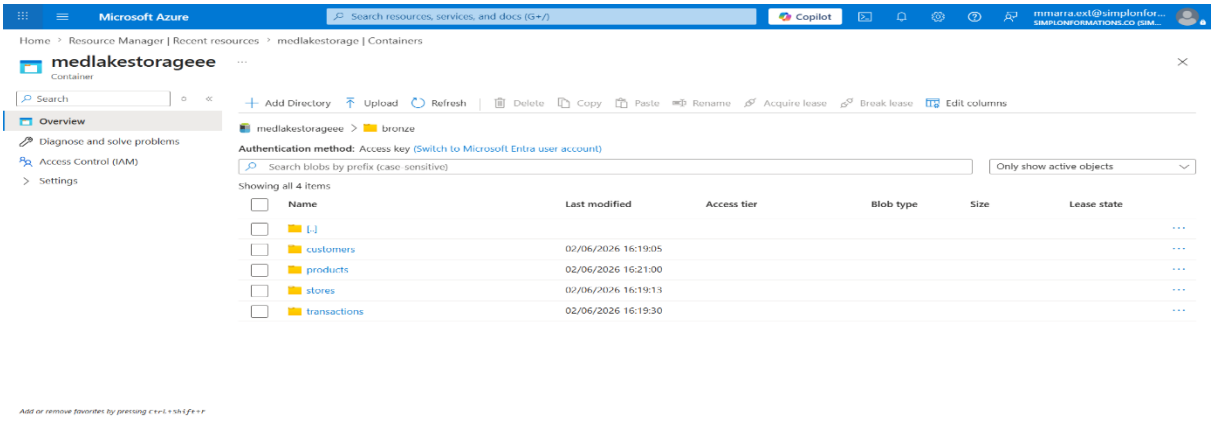
Access between Azure Data Factory and various sources is managed through **Linked Services**. This centralized approach prevents the exposure of passwords or keys within the pipeline code, enhancing overall security governance.

Orchestration Strategy: PL_Ingest_To_Bronze

The ingestion pipeline utilizes a hybrid execution strategy to maximize Data Integration Units (DIU) and reduce latency:

- **Parallel Execution for Dimensions:** Ingestion activities for Copy_Products, Copy_Stores, and Copy_Customers are triggered simultaneously.
- **Dependency Constraints for Facts:** The Copy_Transactions activity is subject to a strict success condition. It only executes after the three dimension activities are successfully completed. This ensures referential integrity, preventing the ingestion of millions of transactions before the necessary reference data is available.





Phase II: Cleansing and Normalization (Silver Layer)

Once raw data is stored in ADLS Gen2, Databricks is employed to transform the Bronze data into a "Single Source of Truth."

Data Quality Operations

PySpark scripts are utilized to apply rigorous quality rules:

- **Deduplication:** The .dropDuplicates() method is used to eliminate any redundancy introduced during ingestion.
- **Null Value Management:** Records missing critical identifiers (e.g., a transaction without a Customer ID) are removed using .dropna() to ensure healthy joins.
- **Type Casting:** Fields are strictly converted, such as transforming string-based dates into formal DateType formats.

Governance and Schema Enforcement

The data is saved in Parquet/Delta format. A critical feature implemented is **Schema Enforcement** via the overwriteSchema option. This validates and dynamically updates table structures, preventing data corruption from unanticipated schema changes in the source systems.

Phase III: Business Intelligence and Aggregation (Gold Layer)

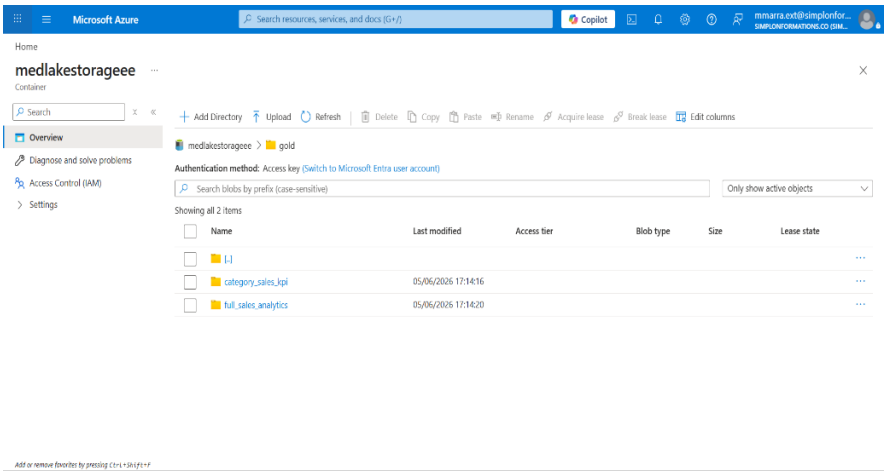
The Gold layer represents the final stage where technical engineering meets business requirements. Data is refined into specific analytical use cases.

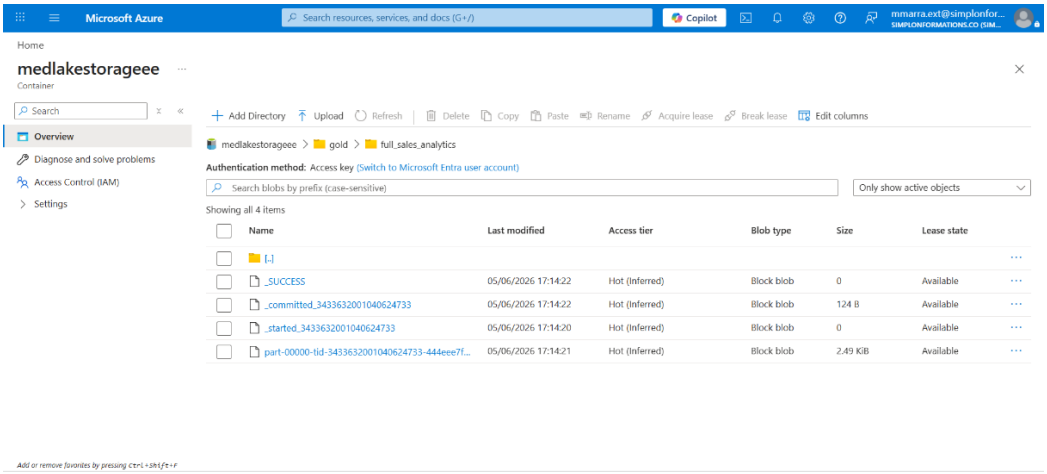
KPI Construction

Using PySpark joins (.join(..., "left")), transaction data is enriched with product metadata. New metrics are calculated "on the fly," such as **Revenue** (Quantity × Price). Aggregation functions (groupBy, sum) are then applied to generate category-specific sales reports.

OBT (One Big Table) Modeling

To optimize performance for BI tools like Power BI, the architecture generates "denormalized" flat tables. By pre-calculating complex joins across millions of rows within Databricks' distributed environment, the system provides a fluid user experience and avoids heavy compute loads during dashboard refreshes.





Conclusions and Evolution Roadmap

The implemented Medallion Architecture provides three primary benefits:

1. **Scalability:** Independent scaling of storage (ADLS) and compute (Databricks) allows for resource optimization based on demand.
2. **Resilience:** Advanced error management in ADF and the preservation of raw data in the Bronze zone allow for easy recovery from failures.
3. **Performance:** Parallel ingestion and PySpark's distributed processing significantly reduce the "Time-to-Insight."

Future Enhancements

While production-ready, the document identifies two key areas for future iteration:

- **Security:** Transitioning from Access Keys to **Azure Managed Identities** to achieve a "Zero-Trust" security posture.
- **Velocity:** Moving from batch processing to continuous streams using **Spark Structured Streaming** for near real-time analytical capabilities.